National Aeronautics and
Space Administration

Interpretable and generative models for mouse spaceflight

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Biological & **P**hysical **S**ciences

PROJECT

One dataset, two machine-learning questions

We used mouse bulk RNA-seq to study how spaceflight changes genes, pathways, and tissue state.

NASA OSDR API Mus musculus bulk RNA-seq | spaceflight (FLT) and ground control (GC) | multiple missions and tissues

01 Which biological programs change?

expiMap

Train a pathway-constrained reference on ARCHS4, then map OSDR samples into mouse Reactome programs.

Output: tissue-specific pathway shifts

02 Can synthetic profiles sharpen the analysis?

Conditional generation

Generate matched expression profiles by tissue, study context, and FLT or GC condition.

Output: validated profiles and gene ranking

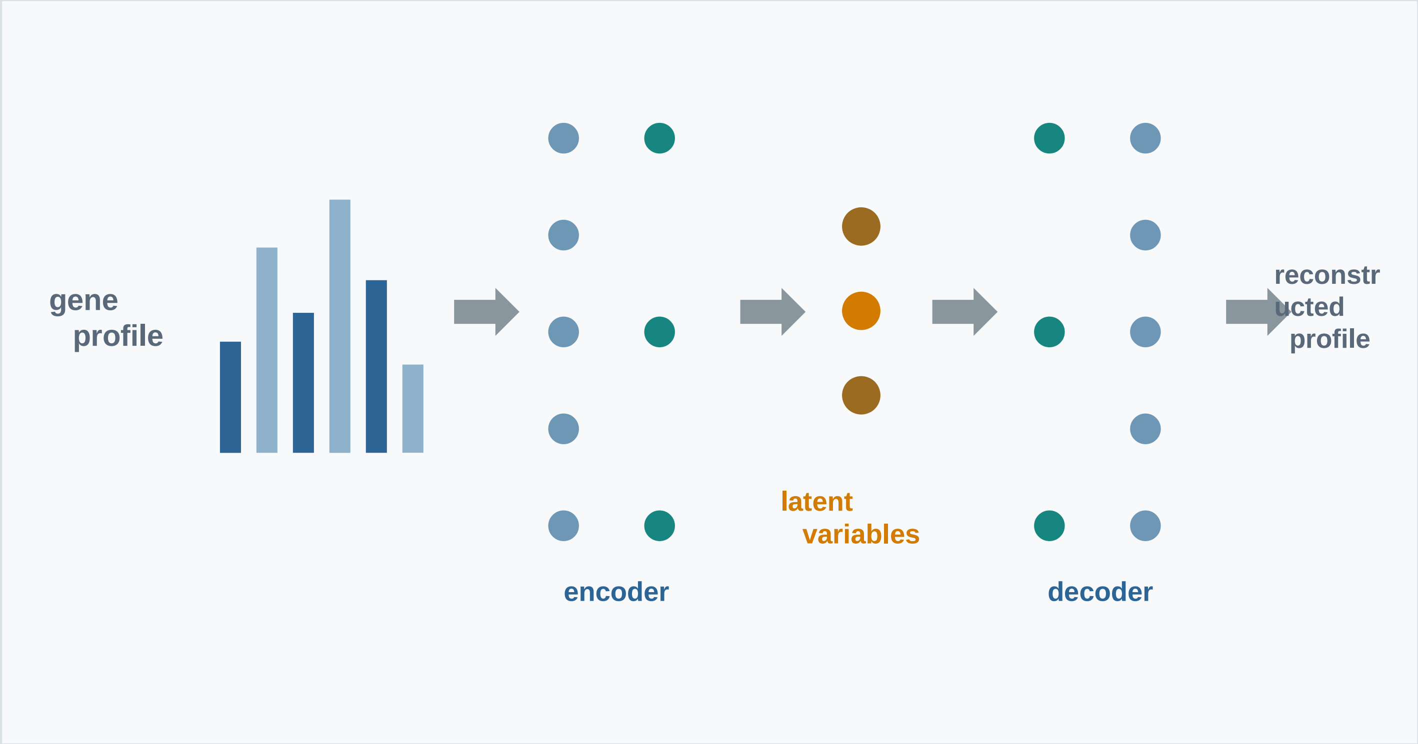
Data source: NASA Open Science Data Repository Biological Data API.

NEURAL NETWORKS

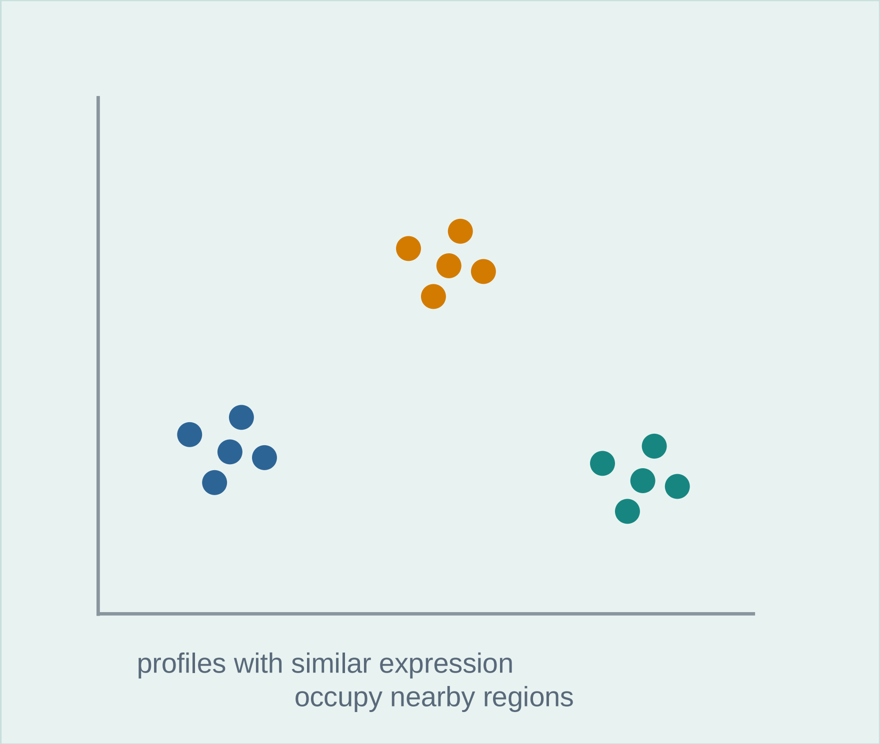
Autoencoders compress expression into a latent space

The encoder summarizes a gene profile; the decoder learns enough structure to reconstruct it.

AUTOENCODER



LATENT SPACE



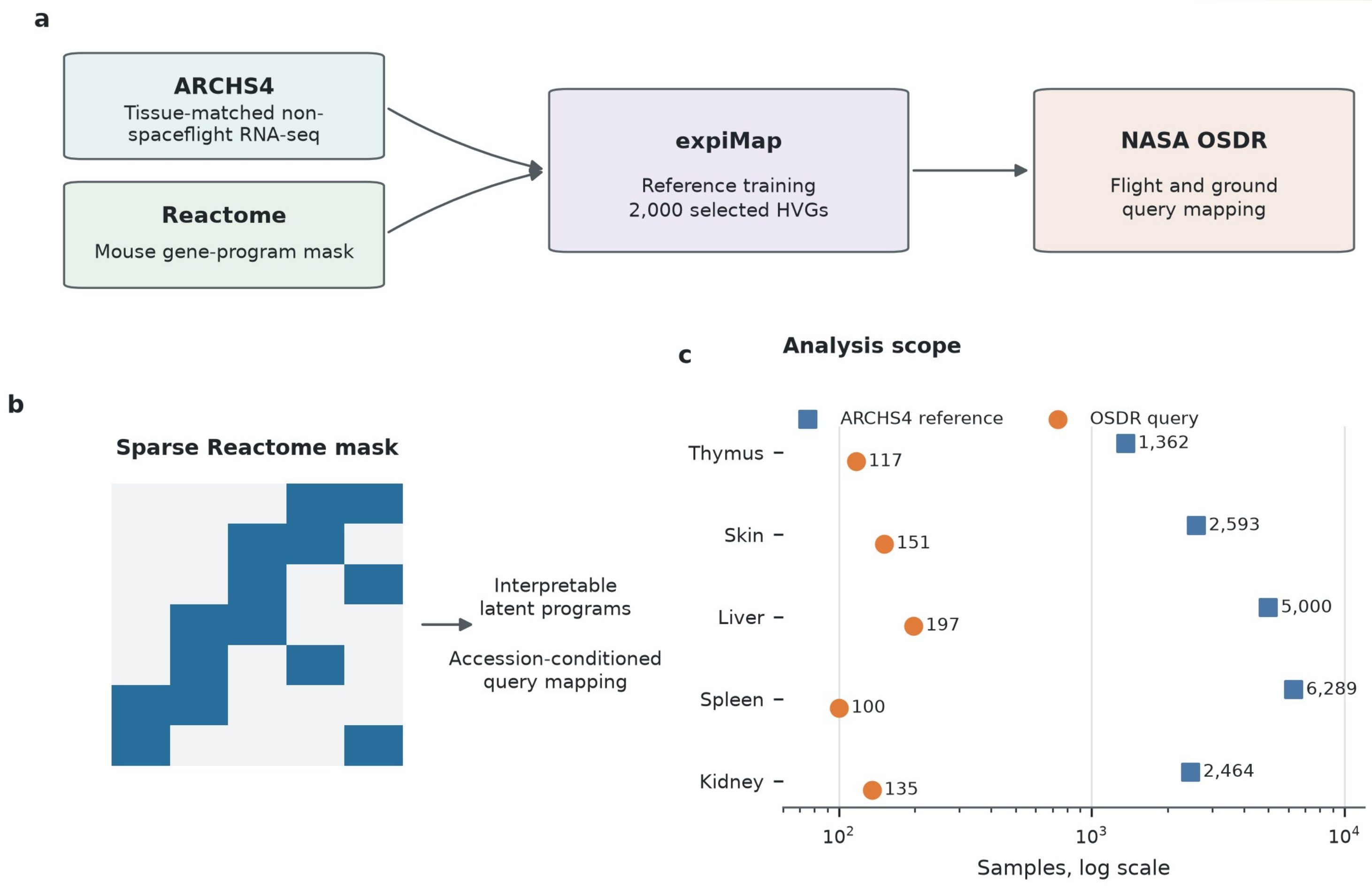
expiMap adds interpretation: each constrained latent variable corresponds to a Reactome gene program.

Concept adapted from the midpoint presentation. expiMap: Lotfollahi et al., Nature Cell Biology (2023).

EXPIMAP

Reactome pathways make the latent space interpretable

Tissue-matched ARCHS4 references define the background; OSDR FLT and GC samples are mapped as queries.



MODEL SCOPE

17,708 ARCHS4 reference profiles

700 OSDR profiles in the primary effects

~2,000 highly variable genes per tissue

319-387 Reactome programs per primary model

Reference training and query mapping were run separately for each tissue.

Source: expiMap manuscript Figure 1 and Tables 1/S25. Implementation: src/expiMap_scarches/nasa_mouse_expimap/.

EXPIMAP RESULTS

Four tissues produced the clearest pathway patterns

Effects were estimated within each project before the project-level shifts were combined.

Thymus

117 samples | 5 projects

LOWER IN FLIGHT

DNA repair; RHOA cytoskeletal cycle; lymphoid-stromal interactions

Known involution may also involve lower repair, motility, and niche coordination.

Skin

151 samples | 4 projects

LOWER IN FLIGHT

Chromatin regulation; DNA repair; Hedgehog; sphingolipids; cell junctions

Barrier injury may include a broader loss of tissue maintenance and coordination.

Liver

197 samples | 9 projects

LOWER IN FLIGHT

MHC class II antigen presentation; T-cell receptor signaling

Metabolic heterogeneity coexisted with lower adaptive immune communication.

Spleen

100 samples | 5 projects

LOWER IN FLIGHT

T-cell receptor signaling; neutrophil degranulation; C-type lectin signaling

The most consistent multi-pathway result joined adaptive and innate immune changes.

Source: expiMap manuscript Figures 3 and 6. Lower latent scores do not by themselves prove biochemical inhibition.

EXPIMAP EVIDENCE

Cross-checks narrowed the biological interpretation

Blue cells mark directional support. Conventional GSEA FDR is reported separately.

Tissue and pathway	ssGSEA	GSEA	Held-out	3 seeds	Composition	GSEA FDR
Thymus: DNA repair	+	+	+	+	+	**<0.001**
Thymus: Lymphoid-stromal interactions	-	-	+	+	+	1.000
Thymus: RHOA cytoskeletal cycle	+	+	+	+	+	0.154
Skin: Cell-cell junction organization	-	+	+	+	+	**0.007**
Skin: Chromatin-modifying enzymes	+	+	+	+	+	**<0.001**
Skin: DNA repair	+	+	+	+	+	**<0.001**
Skin: Hedgehog signaling	+	+	+	+	+	0.280
Skin: Sphingolipid metabolism	+	+	+	+	+	0.897
Liver: MHC class II antigen presentation	+	+	+	+	+	0.121
Liver: T-cell receptor signaling	+	+	+	+	+	0.051
Spleen: C-type lectin receptor signaling	+	+	+	+	+	**0.017**
Spleen: Neutrophil degranulation program	+	+	+	+	+	**<0.001**
Spleen: T-cell receptor signaling	+	+	+	+	+	**0.042**

WHAT REMAINED

Spleen All three programs passed GSEA FDR < 0.05 and all five directional checks.

Skin Chromatin, DNA repair, and cell-junction programs passed GSEA FDR < 0.05.

Thymus DNA repair passed GSEA FDR < 0.001; the cytoskeletal and niche programs had directional support.

Liver MHC II and T-cell receptor scores were lower, but conventional FDR was 0.121 and 0.051.

Spleen was the strongest multi-pathway result.

Source: expiMap manuscript Table 2 and Table S24. Held-out checks are internal project-wise validation.

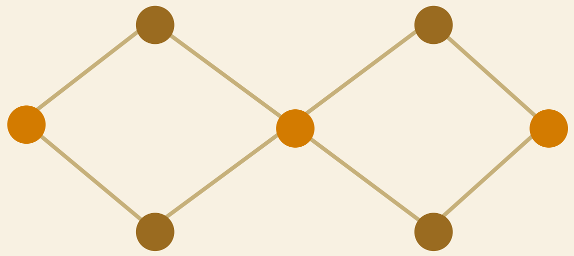
What is synthetic transcriptomics?

A generator learns patterns in measured RNA-seq data, then samples new expression vectors under chosen conditions.

01 Observed profiles

Measured gene expression from real mouse tissue

02 Learn the distribution



Capture gene relationships and variation by tissue and FLT/GC condition

03 Synthetic profiles

Sample new numeric expression vectors for a selected context

QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

Observed spaceflight cohort

1,610 profiles

75 accessions

835 FLT

775 GC

ARCHS4 mouse

Reference pretraining cohort

997,515 profiles screened → **17,244** selected

20 tissues

Study effects can look like spaceflight biology.

01 Preserve tissue and FLT/GC structure.

02 Avoid memorizing individual profiles.

03 Test biological effects in observed OSDR samples.

Sources: NASA OSDR Biological Data API; ARCHS4 mouse v2.5.

METHOD

We built a configurable bulk RNA-seq generation pipeline

The downstream analysis used the outlined options; the remaining choices stay configurable.

01 Data scope

Sources
- **OSDR API**
- **ARCHS4**

Studies
- Single
- **Multiple**

Tissues
- Per tissue
- **All tissues**

02 Transform

Expression
- Raw
- CPM
- **TPM**

Scaling
- None
- Z-score
- **MaxAbs**

Features
- All
- HVG
- **L1000**

03 Harmonization

Global or study correction
- **None**
- Within-study z-score
- ComBat / MBatch
- MOBER

04 Model training

Generator
- WGAN-GP
- **DDIM**

Training source
- OSDR only
- ARCHS4 only
- **Pretrain + adapt**

05 Conditioning

Model inputs
- **Tissue**
- **FLT / GC**
- **Accession**
- **Material type**
- Sex / age

Downstream configuration

PREPROCESS: TPM / MaxAbs / 974 landmarks → PRETRAIN: ARCHS4 → ADAPT: OSDR → GENERATE: Conditional DDIM

CONDITION: Tissue | FLT / GC | Accession | Material type — HARMONIZE: None — SCOPE: All tissues

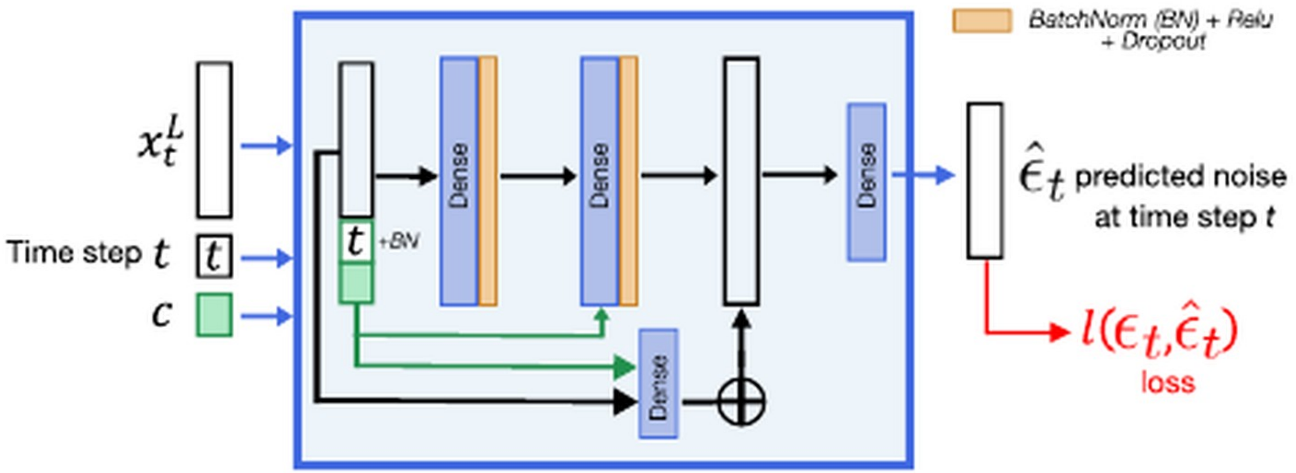
Screen: 463 planned configurations and nine matched liver harmonization arms. Models: Vinas et al. (2022); Lacan et al. (2026).

VALIDATION

DDIM had lower separability and distributional distance

Both generators matched expression. DDIM was harder to distinguish from real profiles and had higher F1.

Architecture from Lacan et al.



Residual dense denoiser conditioned on timestep and tissue.

OSDR adaptation adds FLT/GC, accession and material context through LoRA.

ARCHS4 tissue probe Real 0.781 | Synthetic 0.781

Balanced accuracy is preserved in generated tissue labels.

Metric	WGAN-GP	DDIM	Read
Correlation	0.976	0.974	similar
F1	0.985	0.997	higher
Adversarial acc.	0.636	0.475	near 0.5
FD / real P95	0.144	0.074	lower

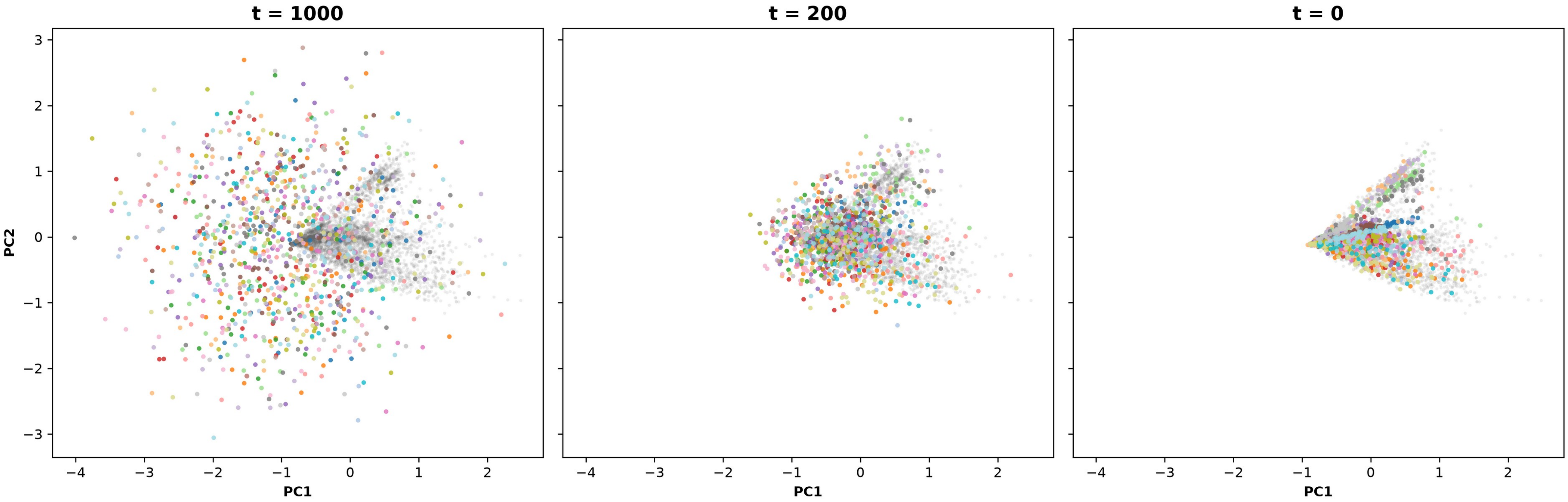
Use DDIM Higher F1 with lower separability and FD.

AA near 0.5 means real and generated profiles are difficult to separate.

Architecture excerpt: Lacan et al. (2026), Fig. 1C, doi:10.1186/s12859-026-06470-8. Metrics use each model's stated split and are not paired.

Diffusion learns tissue structure from noise

The same PCA axes follow 1,024 generated profiles through reverse diffusion.



t = 1000: noise

t = 200: partial structure

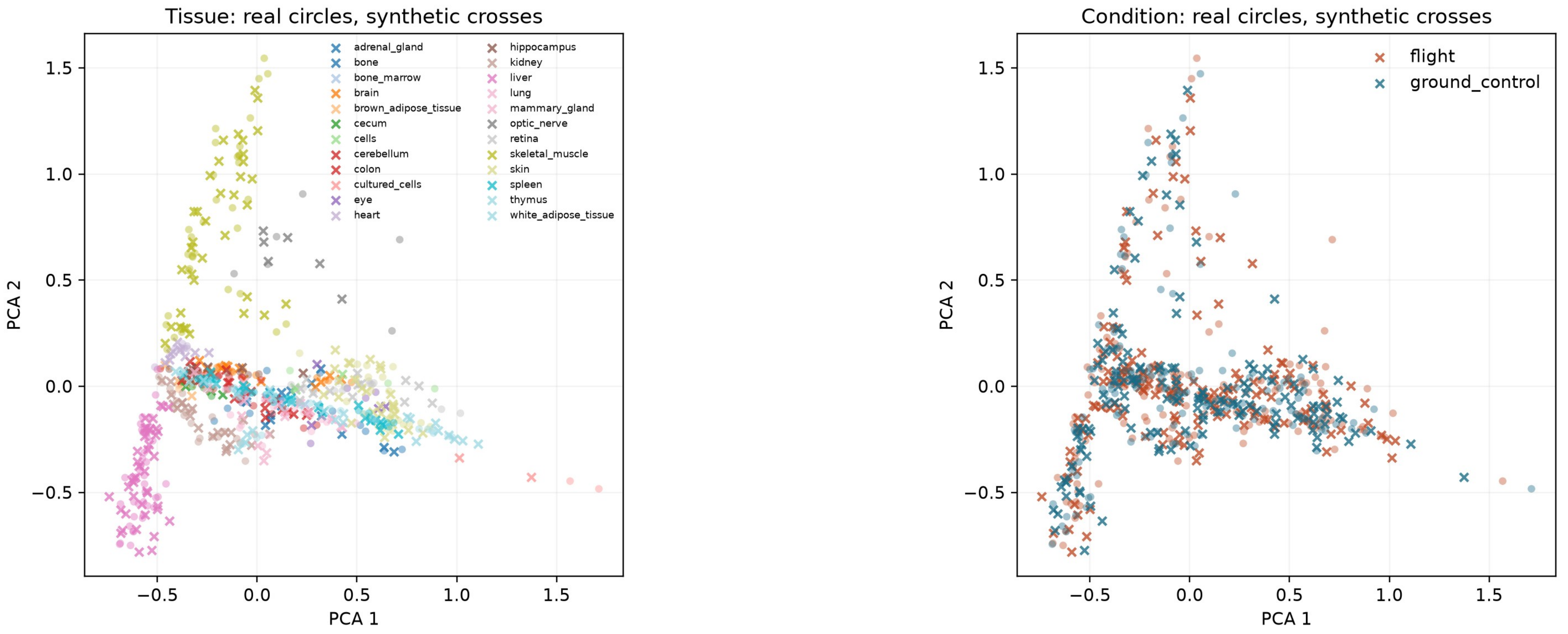
t = 0: tissue-conditioned profiles

Gray points are real ARCHS4 profiles; colors identify generated tissue conditions. PC1 and PC2 limits are shared across panels.

DIFFUSION OUTPUT

Generated profiles track the real OSDR PCA manifold

Circles are locked real OSDR profiles; crosses are matched DDIM profiles from generation seed 5020.



Tissues separate clearly here. Flight and ground-control samples overlap much more, so we test that difference statistically.

PCA was fitted in the common locked OSDR expression space. This is PCA, not UMAP.

Five arms separate gene ranking from classifier fitting

Guided arms use synthetic profiles to help choose genes; they do not treat generated profiles as new animals.

INPUTS R real OSDR S matched DDIM GUIDED R + S help rank genes 5% S carries 5% of total fit weight

ANALYSIS ARM	GENE RANKING	CLASSIFIER FIT	WHAT IT TESTS
Real only	R → Rank from real	R → Fit real	Baseline
Generated only	S → Rank from generated	S → Fit generated	Synthetic-only stress test
Real + generated	R + S → Consensus rank	R + S → Equal total weight	Direct augmentation
Guided: real fit	R + S → Consensus rank	R → Fit real only	Feature guidance only
Guided: 5% synthetic	R + S → Consensus rank	R + S → S = 5% weight	Guidance + light regularization

All five arms → **Held-out real profiles** → **BA | AUROC | AP** → **Choose an eligible arm within each tissue**

Association test FLT vs GC effects and BH FDR use real OSDR profiles only. **Animal n stays unchanged.**

ANALYSIS

Consensus ranking chooses the classifier input genes

Real and generated rankings combine, then only the top k genes are used for FLT-GC classification.

1 REAL RANKING

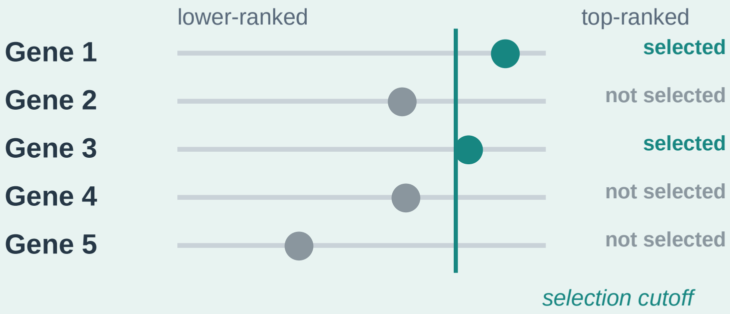
Rank all 974 genes using real OSDR

2 GENERATED RANKING

Rank the same 974 genes using generated profiles

3 CONSENSUS RANKING

Genes supported by both move upward



Top-ranked genes become classifier inputs

Rank 974 genes

real, generated, or consensus

Keep the top k

k = 10, 25, 50, or 100

validation chooses k

Fit FLT-GC classifier

using only the selected gene columns

Five genes are schematic examples from the 974-gene ranking. Real and generated normalized ranks are combined.

ANALYSIS

Synthetic guidance can shift the FLT/GC boundary

Opaque points train the classifier; transparent points are the same held-out real profiles in both panels.

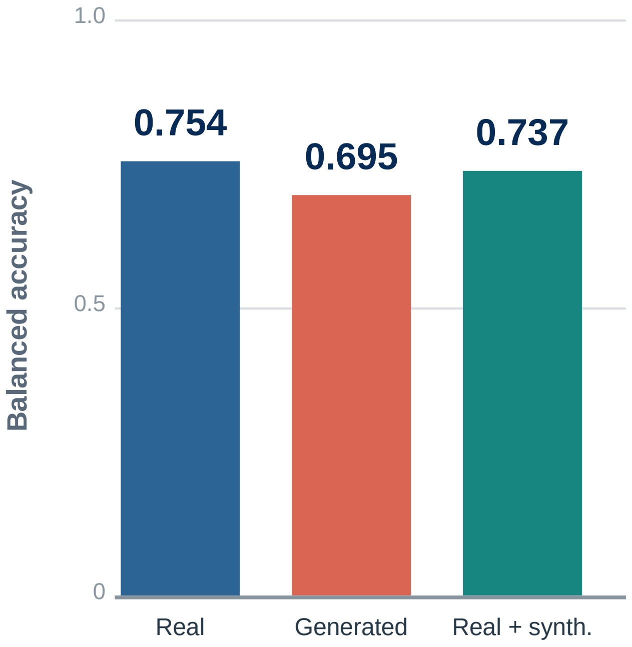
Schematic of the held-out evaluation used for each tissue-specific classifier candidate.

UTILITY

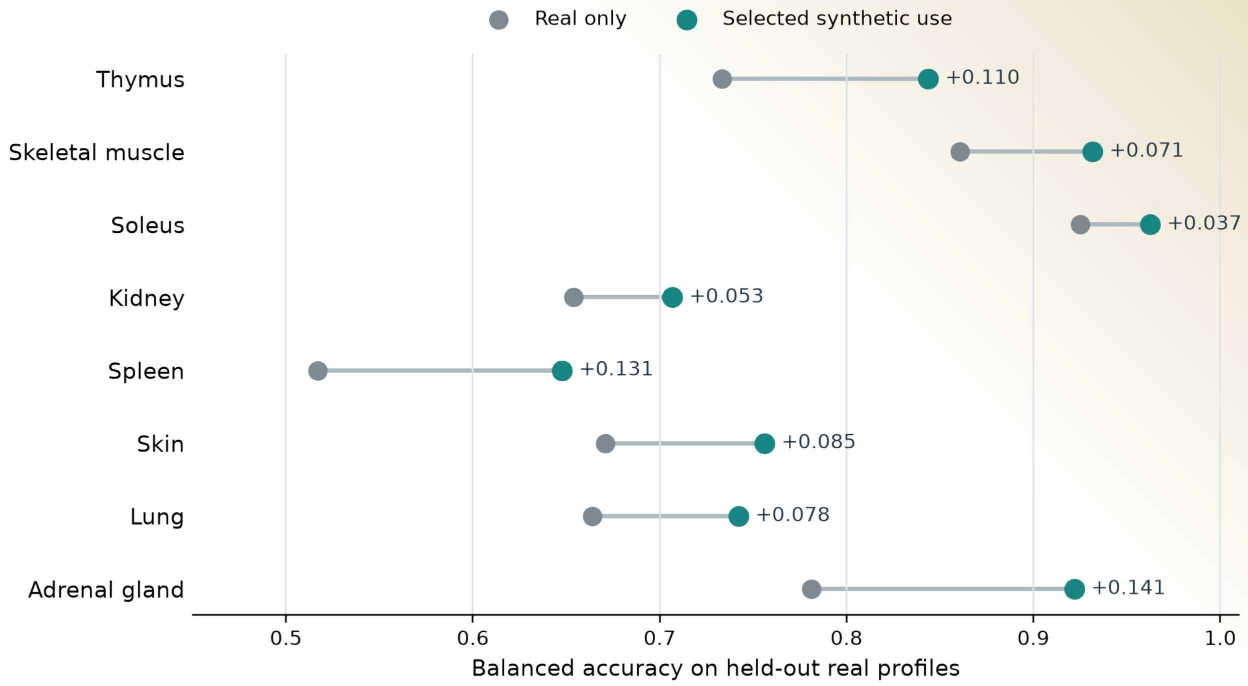
Pooled training missed improvements seen within tissues

The pooled FLT/GC classifier declined, while several tissue-specific classifiers improved.

Pooled across tissues



Selected use within each tissue



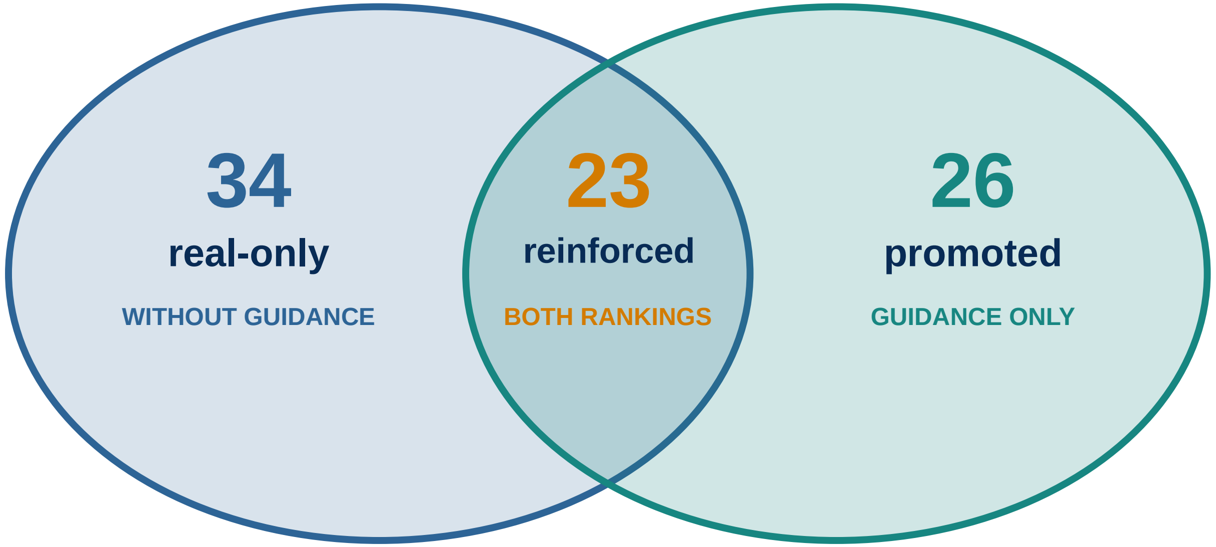
The useful arm differed by tissue.

Synthetic guidance changed ranking, not statistical evidence

Blue marks real-only selection; teal marks synthetic-guided selection. Every displayed association has real OSDR support.

Stable selection after real-data support

- Blue: stable in real-only ranking
- Teal: stable in synthetic-guided ranking



Real OSDR evidence gate

- Estimate FLT vs GC inside each accession
- Combine accession effects with a random-effects model
- Apply BH FDR within each tissue

23 reinforced + 26 promoted

49 synthetic-informed associations

The 34 real-only associations remain real-data findings; synthetic guidance did not reinforce them.

Counts are tissue-gene associations after BH FDR and compatible real-effect direction; genes may recur across tissues.

Selection and literature are separate dimensions

All 49 synthetic-informed associations received both a selection status and a literature interpretation.

01 Fix the candidate
Gene, tissue, and FLT direction

02 Search
Spaceflight and mechanism literature

03 Compare
Observed result with published evidence

04 Record
Category, rationale, and source IDs

Selection status

26 Promoted
Stable only with synthetic guidance

23 Reinforced
Stable with and without guidance

Literature interpretation

Class	Count	Meaning
Aligning	22	Direct or same-tissue process agreement
Complementary	19	Related process or mechanism
Ambiguous	4	Mixed or context-dependent evidence
Unmatched	4	No sufficiently specific match found

A reinforced gene can be complementary, ambiguous, or unmatched; an aligning gene can be promoted.

Table S16 records gene-level rationale, evidence scope and source IDs; Table S17 provides 33 citations with DOI/URL and data relationship.

ANALYSIS COVERAGE

The screen covered all 27 completed tissue analyses

The biological discussion narrows only after reporting synthetic-informed, real-only, and null outcomes.

10 Synthetic-informed

Promoted or reinforced BH-FDR gene

ANALYSIS UNITS

- Adrenal Gland
- Eye
- Kidney
- Skeletal muscle (pooled)
- Skin
- Spleen
- Thymus
- Gastrocnemius
- Soleus
- Tibialis Anterior

5 Real BH-FDR only

Real association, no synthetic-informed selection

ANALYSIS UNITS

- Heart
- Liver
- Retina
- EDL
- Quadriceps

12 No BH-FDR gene

No BH-FDR gene in the 974-gene panel

ANALYSIS UNITS

- Bone
- Bone Marrow
- Brain
- Brown Adipose Tissue
- Cecum
- Cerebellum
- Colon
- Hippocampus
- Lung
- Mammary Gland
- Optic Nerve
- White Adipose Tissue

Twenty-two canonical tissues plus EDL, gastrocnemius, quadriceps, soleus and tibialis anterior; complete counts in Table S12.

GENE INVENTORY

Ten tissue analyses contained synthetic-informed genes

All 49 associations passed BH FDR in real OSDR data; synthetic profiles affected prioritization, not the statistical test.

Rows: FLT direction + status — aligning — complementary — ambiguous — unmatched

Gene color = literature | Non-empty rows shown

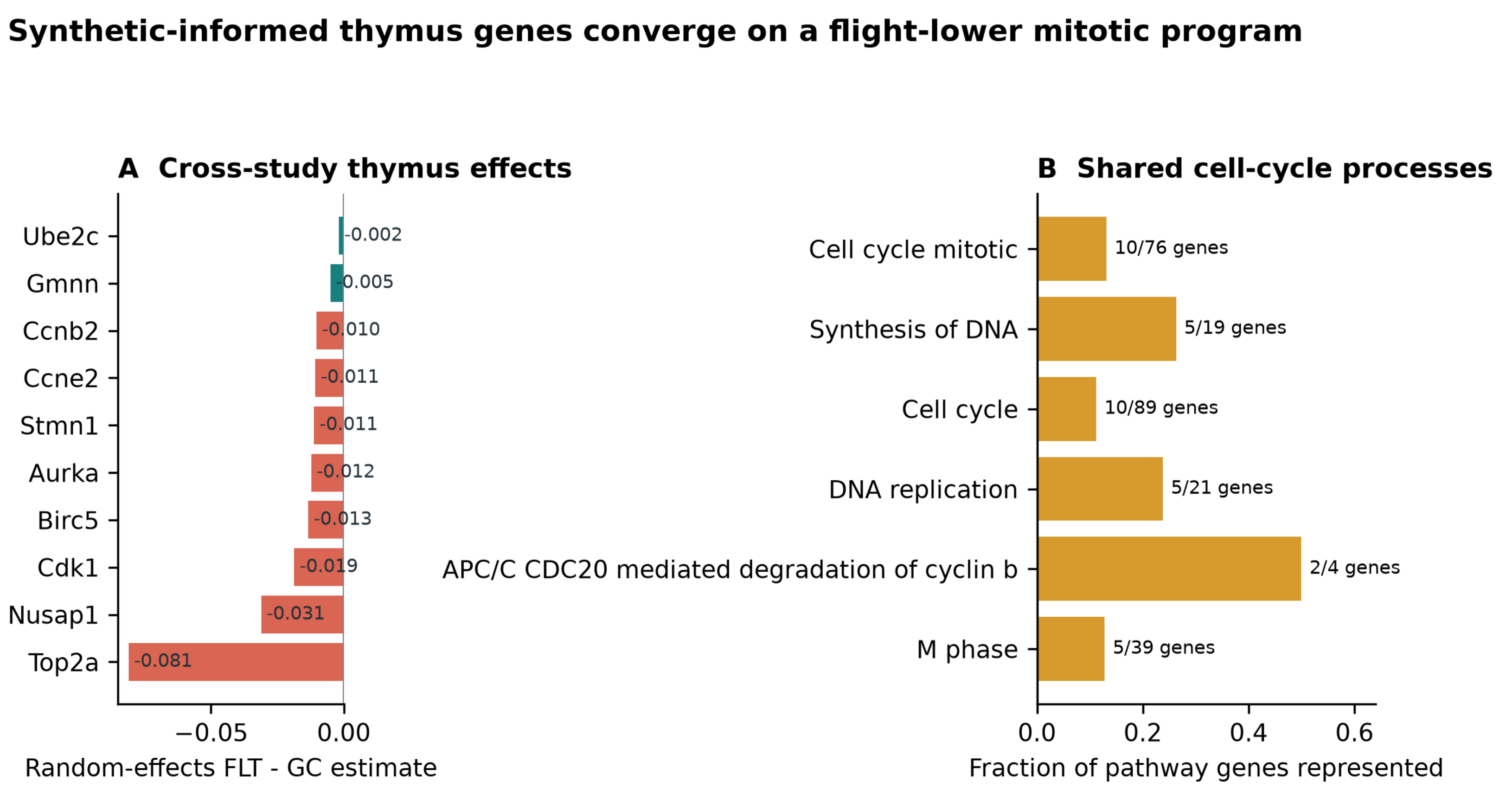
Thymus 16 genes	FLT higher	Promoted	*Hsd17b11*, *Etv1*
	FLT higher	Reinforced	*Snx7*
	FLT lower	Promoted	*Nusap1*, *Stmn1*, *Birc5*, *Cdk1*, *Top2a*, *Ccnb2*, *Aurka*, *Ccne2*, *Kif20a*, *Pcna*, *Ccnf*
	FLT lower	Reinforced	*Ube2c*, *Gmnn*
Spleen 4 genes	FLT higher	Promoted	*Rai14*, *Myl9*, *Ptprk*
	FLT higher	Reinforced	*Loxl1*
Kidney 2 genes	FLT higher	Promoted	*Inpp4b*
	FLT higher	Reinforced	*Slc37a4*
Gastrocnemius 2 genes	FLT higher	Promoted	*Nfkbia*
	FLT lower	Promoted	*Fhl2*
Eye 1 gene	FLT lower	Reinforced	*Klhl21*
Skeletal muscle (pooled) 12 genes	FLT higher	Reinforced	*Sox4*, *Cebpd*, *Sh3bp5*, *Prkcd*, *Arid5b*, *Sesn1*, *Tle1*
	FLT lower	Promoted	*Klhl21*, *Mapkapk5*, *Reep5*, *Itgb5*
	FLT lower	Reinforced	*Bphl*
Soleus 5 genes	FLT higher	Reinforced	*Tpm1*
	FLT lower	Reinforced	*Bdh1*, *Ech1*, *Bnip3*, *Decr1*
Tibialis anterior 4 genes	FLT higher	Promoted	*Cebpd*
	FLT higher	Reinforced	*Cdkn1a*, *St3gal5*, *Bnip3*
Adrenal gland 2 genes	FLT lower	Promoted	*Psmb8*
	FLT lower	Reinforced	*Tspan4*
Skin 1 gene	FLT higher	Promoted	*Plscr1*

Separate rows report FLT direction and selection status; gene color independently reports literature interpretation for all 49 associations.

THYMUS

Thymus points to lower proliferative renewal

The cell-cycle program persisted without material conditioning; two FLT-higher genes extend the hypothesis.



16 synthetic-informed BH-FDR associations

13 promoted | 3 reinforced

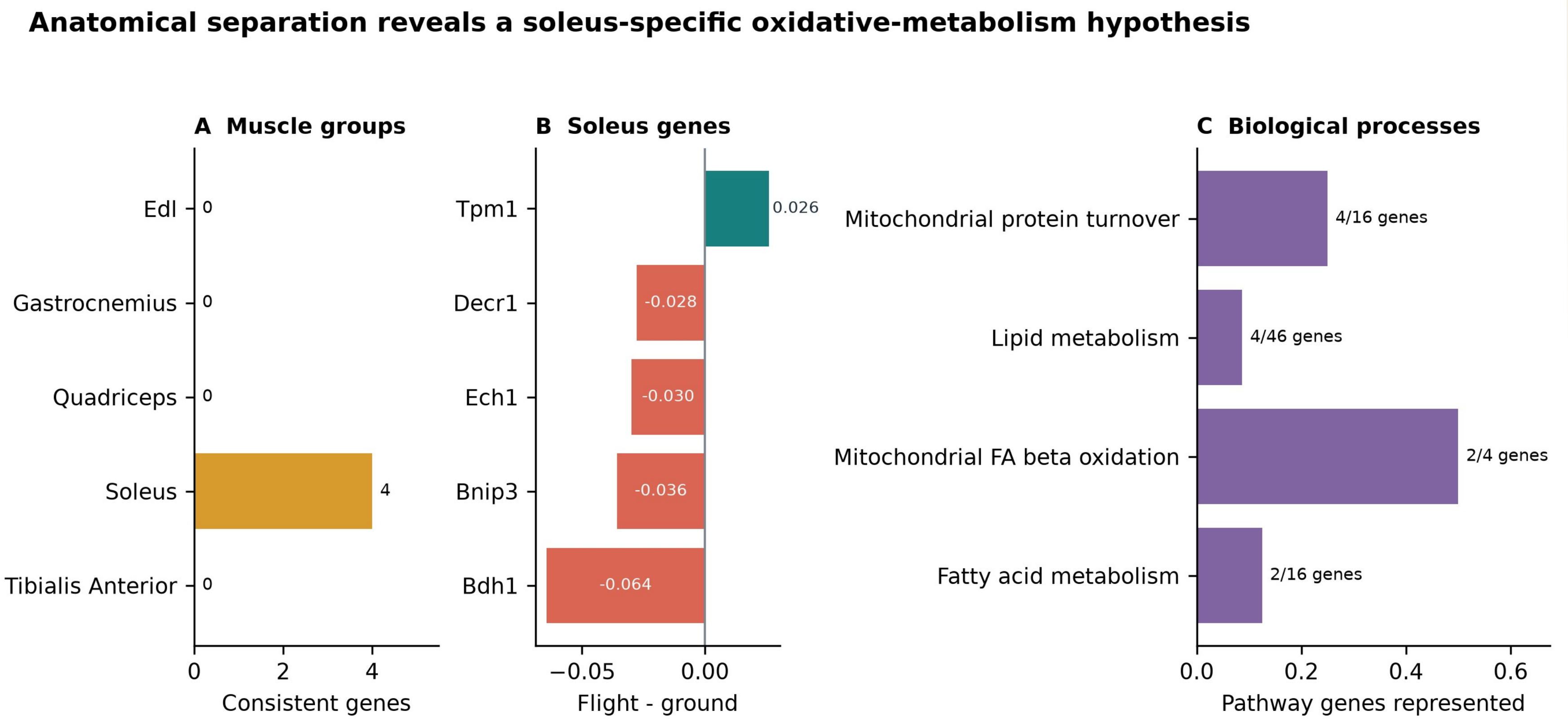
- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

Context: Gridley et al. (2013), Horie et al. (2019), Keenan et al. (2025), and Shi et al. (2026).

SOLEUS

Soleus reinforces a mitochondrial and lipid program

The real-data program is coherent, but its synthetic reinforcement required explicit material conditioning.



0.925 to 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

Literature context: Gambara et al. (2017) and Stein et al. (2002).

Additional tissues produced distinct hypotheses

Promoted and reinforced selections are shown separately for each tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Pooled muscle	Promoted	Lower: Klhl21, Mapkapk5, Reep5, Itgb5	Heterogeneous stress, differentiation, interferon and sialic-acid response; interpret with the anatomical muscle groups.
	Reinforced	Higher: Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1; lower: Bphl	
Kidney	Promoted	Inpp4b higher	Renal phosphoinositide signaling and glucose-handling hypothesis; the pair had no shared Reactome enrichment.
	Reinforced	Slc37a4 higher	
Spleen	Promoted	Rai14, Myl9, Ptprk higher	Adhesion, actomyosin and extracellular-matrix or immune-organization hypothesis; no coherent pathway enrichment.
	Reinforced	Loxl1 higher	
Skin	Promoted	Plscr1 higher	Interferon-linked skin candidate within broader cell-cycle and DNA-repair responses; a single-gene result.
	Reinforced	None	

All listed effects passed BH FDR in observed OSDR profiles; 'None' means no gene in that selection category.

ADDITIONAL FINDINGS II

Eye, adrenal and muscle-group results remain tissue-specific

Promoted and reinforced selections are separated within every tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Eye	Promoted	None	Process-level alignment with lower proliferation and cytokinesis in flight eye tissue; not an exact prior gene replication.
	Reinforced	Klhl21 lower	
Adrenal gland	Promoted	Psmb8 lower	Literature-unmatched immune, proteostasis or tissue-composition candidates; no adrenal mechanism is established.
	Reinforced	Tspan4 lower	
Gastrocnemius	Promoted	Nfkbia higher; Fhl2 lower	NF-kappaB stress alignment plus an autophagy or myogenesis candidate; the two genes do not form a coherent pathway.
	Reinforced	None	
Tibialis anterior	Promoted	Cebpd higher	Stress, cell-cycle, ganglioside-signaling and mitophagy candidates with mixed or complementary prior evidence.
	Reinforced	Cdkn1a, St3gal5, Bnip3 higher	

Interpretations follow the independent literature annotations in Table S16; 'None' means no gene in that selection category.

TAKEAWAYS

Synthetic data worked best as a tissue-specific prior

The generator helped prioritize observed signal. Biological claims still come from OSDR profiles.

01 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

02 Use

Tissue-specific consensus ranking and light synthetic regularization improved held-out prediction in selected tissues.

03 Interpret

Promoted and reinforced genes supported tissue-specific hypotheses. Literature review separated prior alignment from complementary findings.

Thank you

Questions?

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Compute	NASA Ames Research Center
Code and manuscript	github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.